# SQL Comprehensive Practice Question

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**Database Context:**
You are managing a **Library Management System**. Below are the tables you need to create:

**Tables & Structure:**

1. **Books**
   - BookID (INT, Primary Key)
   - Title (VARCHAR(100))
   - Author (VARCHAR(100))
   - Genre (VARCHAR(50))
   - PublishedYear (INT)
   - Price (DECIMAL(6,2))
2. **Members**
   - MemberID (INT, Primary Key)
   - FullName (VARCHAR(100))
   - Address (VARCHAR(100))
   - Gender (VARCHAR(10))
   - JoinDate (DATE)
3. **Loans**
   - LoanID (INT, Primary Key)
   - BookID (INT, Foreign Key referencing Books(BookID))
   - MemberID (INT, Foreign Key referencing Members(MemberID))
   - LoanDate (DATE)
   - ReturnDate (DATE)
4. **Staff**
   - StaffID (INT, Primary Key)
   - Name (VARCHAR(100))
   - Position (VARCHAR(50))
   - Salary (DECIMAL(7,2))
   - Branch (VARCHAR(50))

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## Practice Questions:

### 1. Basic SQL Operations

1. Create the **Books**, **Members**, **Loans**, and **Staff** tables with appropriate constraints.
2. Insert **5 records** into each table.
3. Display all books authored by **"J.K. Rowling"**.
4. Display all members who joined in the year **2023**.

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## 2. Aggregate Functions & Grouping

5. Find the **total number of books** in each genre.
6. Calculate the **average price** of books published after **2015**.
7. Count the **total number of male and female members**.
8. Calculate the **total salary** of staff members working in the **"Downtown" branch**.
9. Find the **total number of loans** issued per year.

## 3. Sorting & Ranking

10. Display the **top 3 most expensive books**.
11. Find the **second highest salary** among staff.
12. List all books **sorted by title**.
13. Sort all members **by JoinDate** in descending order.

## 4. Joins & Relationships

14. Display the **titles of books borrowed by "John Doe"**.
15. Find the **names of staff** who work in the same branch as **"Downtown"**.
16. Retrieve the **full details of members who have borrowed 'Harry Potter'**.
17. Display the **books borrowed by members** who joined in **2023**.

## 5. Subqueries

18. Find the **highest salary among staff** in the **"Uptown" branch**.
19. Retrieve the **number of books** that have never been borrowed.
20. Find the **genre of books** that have the **highest average price**.
21. List **members who have borrowed more books than the average number**.

## 6. Data Modification

22. Delete all books published **before 2000**.
23. Update the **salary of staff** in the "Downtown" branch by **10%**.
24. Mark all books borrowed by **'John Doe'** as returned by setting **ReturnDate** to today.
25. Add **5%** to the price of all books in the **"Fantasy" genre**.

## 7. Advanced SQL Concepts

26. Create a **view** showing the **books borrowed in 2024** along with borrower names.
27. Create a **stored procedure** that accepts a **MemberID** and returns all the books they have borrowed.
28. Create a **trigger** to automatically update the **ReturnDate** to today when a book is returned.

## 8. Bonus Challenges

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29. Write a query to find **members who have borrowed at least 3 different genres** of books.
30. Calculate the **total revenue generated** from loans (assume each loan generates revenue equal to the book's price).

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